Word Codes (Set B)

Year 4 · **Lesson Worksheet**

Worked Examples

Example 1 — Decoding with a Shift: In a code, every letter is shifted forward by 2 (A→C, B→D...). What does the code word 'FQI' decode to?

Answer: DOG — shifting each letter BACK by 2: F→D, Q→O, I→G.

Example 2 — Encoding with a Shift: Using the same +2 shift code, encode the word 'CAT'.

Answer: EVC... check: C→E, A→C, T→V, so the code is 'ECV'. (Each letter moves forward 2 places: C→E, A→C, T→V.)

Example 3 — Worked Pair Method: If CAT is coded as DBU, what shift is being used, and how would you code the word DOG?

Answer: The shift is +1 (each letter moves forward one place: C→D, A→B, T→U). Applying +1 to DOG: D→E, O→P, G→H, giving 'EPH'.

Spotting the Code (15 questions)

1. In a code, every letter is shifted forward by 1 (A→B, B→C...). What does "DPH" decode to?
2. Using a +1 shift code, encode the word "CAT".
3. If the code for DOG is EPH (shift +1), what is the code for PIG? A) QJH B) QJI C) RJH D) QKH E) PJH
4. Select ALL that apply: using a shift of +3 (A→D, B→E, C→F...), which of these are correctly encoded? A) CAT → FDW B) DOG → GRJ C) PIG → SLJ D) SUN → VXQ E) BAT → EDW
5. In a code, every letter is shifted BACK by 1 (B→A, C→B...). What does "EBU" decode to?
6. If BAT is coded as CBU (shift +1), what is the code for RUN? A) SVO B) SVN C) TVO D) SUO E) RVO
7. Using a +2 shift code (A→C, B→D...), encode the word "HEN".
8. In a code, every letter is shifted BACK by 2 (C→A, D→B...). What does "KQI" decode to?
9. If the word FOX is coded as GPY (shift +1), decode the word "JBU" using the same shift.
10. Select ALL that apply: using a shift of -1 (B→A, C→B...), which of these are correctly decoded? A) "DPH" → COG B) "UBU" → TAT C) "CBU" → BAT D) "GPY" → FOX E) "KQI" → LPH
11. Using a +1 shift code, encode the word "PIG".
12. If the word CAT is coded as BZS (shift -1), what is the code for BAT? A) AZS B) BZS C) AYS D) ZYS E) AZT
13. In a code, every letter is shifted forward by 2 (A→C, B→D...). What does the code word "LGZ" decode to?
14. Using a +3 shift code, encode the word "CAT".
15. Select ALL that apply: if the shift used is +2 (A→C, B→D...), which of these pairs is correctly matched? A) HEN→JGP B) PIG→RKI C) DOG→FQI D) SUN→UWP E) CAT→EDV

Applying the Code (15 questions)

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1. In a code, CAT is written as ECV. Work out the shift being used, then use it to encode "DOG".
2. In a code, each letter's code is found by shifting forward by 4 places. Decode the word "GSR".
3. A code represents each letter by shifting back 3 places. If "PTFJ" is a coded word, decode it.
4. Select ALL that apply: a code shifts every letter forward by 5. Which of these pairs are correctly matched? A) DOG→ITL B) CAT→HFY C) PIG→UNL D) SUN→XZS E) FOX→KTC
5. In a code, the word BIRD is written as DKTF. Work out the shift, then encode the word "NEST" using the same shift.
6. A code shifts letters BACK by 4 places. Decode the coded word "PEJ".
7. If the word STAR is coded as VWDU (shift +3), decode the word "ZLQG" using the same shift.
8. In a code, each letter is replaced by the letter 5 places further on in the alphabet, wrapping round after Z if needed. Encode the word "WOLF".
9. Select ALL that apply: a code shifts every letter back by 6 places (wrapping round before A if needed). Which of these pairs are correctly matched? A) GOAT→AIUN B) MOON→GIIH C) HORSE→BILMY D) ZEBRA→TYVLU E) FROG→ZLIA
10. In a code, DUCK is written as FWEM. Work out the shift, then decode "JQWUG" using the same shift.
11. A code shifts every letter forward by 6 places. Encode the word "FISH".
12. If the word MOUSE is coded as PRXVH (shift +3), decode the word "ILVK" using the same shift.
13. In a code, each letter is replaced by the letter 7 places back in the alphabet, wrapping round before A if needed. Decode the coded word "UBKW".
14. Select ALL that apply: a code shifts every letter forward by 4 places. Which of these pairs are correctly matched? A) BEAR→FIEV B) SWAN→WAER C) GOAT→KSEX D) LION→PMSR E) DEER→HIIV
15. In a code, the word HORSE is written as KRUVH. Work out the shift, then use it to encode the word "SNAKE".

Codes in Context (15 questions)

1. A puzzle gives the pair TIGER → VKIGT as a worked example. Work out the shift being used, then decode the word "EJCKT" using the same shift.
2. In a code, each letter's position number has 3 added to it, then it's converted back to a letter. What does the letter G become under this rule?
3. Using a +2 shift code (A→C, B→D…), encode the word "LEG". A) NGI B) NGH C) MGI D) NFI E) NHI
4. In a code, every letter is shifted BACK by 3 places (D→A, E→B, F→C…). What does the code word "WKH" decode to?
5. A puzzle gives the pair HEN → KHQ as a worked example. Work out the shift, then encode the word "CAT" using the same shift.
6. In a code, each letter is replaced by the letter that is the same number of places from the END of the alphabet as it is from the START (so A ↔ Z, B ↔ Y, C ↔ X…). What does the letter D become under this rule? A) W B) X C) Y D) V E) U
7. A puzzle gives the pair BAT → DCV as a worked example. Work out the shift, then decode the word "FQI" using the same shift.
8. In a code, vowels (A, E, I, O, U) are shifted forward by 1, but consonants stay the same. Encode the word "DOG" using this rule.

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9. Select ALL that apply: a code shifts consonants back by 1 place but leaves vowels unchanged. Applying this to the word "CAT", which of these are correct? A) C becomes B. B) A stays A. C) T becomes S. D) A becomes Z. E) The coded word is BAS.

10. A puzzle gives the pair CAT → ECV as a worked example. Using the same shift, encode the word "DOG".

11. In a code, each letter is replaced by the letter that is the same number of places from the END of the alphabet as it is from the START (A ↔ Z, B ↔ Y, C ↔ X…). What does the word "CAB" become under this rule? A) XZY B) YZX C) XYZ D) ZXY E) WZY

12. A two-step code first shifts every letter forward by 1 place, then reverses the order of the letters in the word. Apply this to the word "CAT" and give the final 3-letter result.

13. In a code, each letter's position number has 4 added to it, then it's converted back to a letter. What does the letter P become under this rule?

14. A puzzle gives the pair DOG → GRJ as a worked example. Work out the shift, then decode the word "ZDW" using the same shift. A) WAT B) WAS C) VAS D) WAR E) WAU

15. A two-step code first reverses the order of the letters in the word, then shifts every letter forward by 1 place. Apply this to the word "DOG" and give the final 3-letter result.

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